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Heterogeneous Compute Routing: Capacity-Aware Multi-Policy Workload Placement

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Abstract

Production agentic operations route workloads across heterogeneous substrates: on-premises clusters, hyperscaler-managed inference surfaces, sovereign bare-metal capacity, and forward-deployed edge nodes. The naive view of routing in such environments treats workload placement as a load-balancing problem to be solved at the orchestration plane. This paper argues that routing in heterogeneous compute is a capacity-aware multi-policy decision that reasons simultaneously about workload class, governance envelope, substrate availability, cost regime, and sovereignty constraint. Single-policy routers (latency-only, cost-only, region-only, SKU-only, account-only) fail in production agentic operation because the binding constraint at any given moment may originate in any of the five policy dimensions, and a router that optimizes one dimension while ignoring the others is operating outside the federation envelope without recognizing it.

The paper develops a specification of the capacity-aware multi-policy router and shows how it composes with the K3s + service mesh substrate that the Mission-Ready Decalogy's operational plan documents in production. It identifies five structural failure modes of single-policy routing, specifies the multi-policy decision logic, and binds the routing layer to the Skipjack Protocol's enforcement mechanisms. Routing policy is governance-defined and human-owned; routing decisions within policy are agentic.

This is Paper 2 of the *Managing Agentics Ops* series. Paper 1 (Capacity Sourcing) establishes the six-layer substrate; Paper 3 (Capex/Opex Inflection) examines the cost-regime conditions that determine when routing across substrates beats consolidation on any one.

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ABSTRACT

Production agentic operations route workloads across heterogeneous substrates: on-premises clusters, hyperscaler-managed inference surfaces, sovereign bare-metal capacity, and forward-deployed edge nodes. The naive view of routing in such environments treats workload placement as a load-balancing problem to be solved at the orchestration plane. This paper argues that routing in heterogeneous compute is a **capacity-aware multi-policy decision** that reasons simultaneously about workload class, governance envelope, substrate availability, cost regime, and sovereignty constraint. Single-policy routers (latency-only, cost-only, region-only, SKU-only, account-only) fail in production agentic operation because the binding constraint at any given moment may originate in any of the five policy dimensions, and a router that optimizes one dimension while ignoring the others is operating outside the federation envelope without recognizing it.

The paper develops a specification of the capacity-aware multi-policy router and shows how it composes with the K3s + service mesh substrate that the Mission-Ready Decalogy's operational plan (MR-P9) documents in production.¹ It identifies five structural failure modes of single-policy routing, specifies the multi-policy decision logic, and binds the routing layer to the Skipjack Protocol's enforcement mechanisms. Routing policy is governance-defined and human-owned; routing decisions within policy are agentic. The HGC³AE² framework's role ordering — *humans govern, context is curated, AI executes, master stamps last* — is the architectural anchor.

This is Paper 2 of the *Managing Agentics Ops* series. Paper 1 (Capacity Sourcing) establishes the six-layer substrate over which routing operates; the present paper specifies the routing architecture; Paper 3 (Capex/Opex Inflection) examines the cost-regime conditions that determine when routing across substrates beats consolidation on any one.

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1. INTRODUCTION

Agentic systems do not run on a single substrate. The Robo Stack federation documented in MR-P9 — Omen on-premises + AWS managed + OCI sovereign bare-metal — is one concrete instance of the heterogeneous substrate pattern that mission-ready operation requires; other compositions are admissible.² What is invariant across compositions is the routing question: given a workload to execute, which substrate should execute it, with what data-locality constraints, under which governance envelope, and at what cost regime?

The routing question has historically been answered, in cloud-native environments, by load balancers and scheduler extensions optimizing for a single dominant signal. Latency-aware

1 Forthcoming MR-P9 *Operational Plan* (v1.0-preprint, in concurrent drafting; yks-build#99).

2 Forthcoming MR-P9 *Operational Plan* (v1.0-preprint, in concurrent drafting; yks-build#99).



scheduling routes to whichever endpoint is fastest. Cost-aware scheduling routes to whichever endpoint is cheapest per unit. Region-aware scheduling routes by geographic proximity. SKU-aware scheduling routes by accelerator class. Account-aware scheduling routes across cloud accounts to absorb regional quota exhaustion.³⁴ Each of these works in steady state for the workload class it was designed around. None of them, in isolation, works for the workload class that mission-ready agentic operation produces — a mix of steady-state inference, governance overhead, persistent memory, and burst training that the federation must service across substrates with materially different operational properties.

The argument of this paper is that routing in heterogeneous compute is a **multi-policy decision** that must reason over all five dimensions simultaneously, anchored by a governance-defined envelope that the routing layer cannot exceed. Section 2 identifies five failure modes of single-policy routing observed in production. Section 3 specifies the capacity-aware multi-policy router. Section 4 treats routing as a continuous operational process. Section 5 couples the router to Skipjack enforcement. Section 6 returns the analysis to HGC³AE² and to the MAO Decalogy's operational implications.

2. PROBLEM SET: FIVE FAILURE MODES OF SINGLE-POLICY ROUTING

Single-policy routers fail in production agentic operation in five structurally distinct ways. The failures are observed at every hyperscaler and at every reasonable on-premises substrate; they reflect the mismatch between the routing problem's actual structure and the optimization problem any single-policy router can solve.

2.1 LATENCY-ONLY ROUTING INTO SOVEREIGNTY VIOLATION

A latency-only router selects whichever endpoint responds fastest. In mission-ready agentic operation, the fastest endpoint may sit in a sovereignty regime that the workload was not authorized to operate in — a residency-constrained data class routed to a cross-jurisdiction substrate; an export-controlled workload routed to a region under embargo; a coalition-restricted operation routed outside the coalition perimeter. The router has optimized correctly for its single dimension; the federation has nonetheless executed a workload outside its envelope. The failure is structural: the router lacks the policy dimensions required to refuse the route.⁵

2.2 COST-ONLY ROUTING INTO CAPACITY QUALITY DEGRADATION

A cost-only router selects whichever endpoint is cheapest per unit. The cheapest endpoint at any given moment may be the one carrying silently degraded capacity quality (MAO-P1 §2.2): hardware refresh, thermal throttling, or pool rebalancing has shifted the substrate underneath the SKU without changing the published specifications. The router routes correctly against the price signal; the workload executes at degraded throughput, the comprehension instrument's

3 Klaus Rausch, Waldemar Hummer, Vinod Muthusamy, Alexander Rashed, and Schahram Dustdar, "Towards a Serverless Platform for Edge AI," *IEEE Workshop on Hot Topics in Edge Computing (HotEdge '19)*, 2019, <https://www.usenix.org/conference/hotedge19/presentation/rausch>.

4 Wei Chen et al., "Kubernetes Scheduler Extensions: A Survey," *Future Generation Computer Systems* 137 (2022): 156–172, <https://doi.org/10.1016/j.future.2022.07.007>.

5 Anu Bradford, *The Brussels Effect: How the European Union Rules the World* (Oxford: Oxford University Press, 2020).



calibration signals diverge from their training-time baseline, and the federation's confidence in its own state erodes silently.

2.3 REGION-ONLY ROUTING INTO ACCELERATOR-CLASS MISMATCH

A region-only router selects whichever region is geographically proximal. In 2026, accelerator allocation is not regionally uniform; the closest region may not have the accelerator class the workload requires.⁶ The router's geographic optimization produces correct routing against its single dimension while the workload runs on a sub-optimal accelerator class. Over a long enough operating window, the cumulative cost of misaccelerated workload exceeds the latency cost the regional proximity was meant to optimize.

2.4 SKU-ONLY ROUTING INTO UPSTREAM ALLOCATION COLLAPSE

An SKU-only router selects whichever account currently has the requested SKU available. When upstream allocation tightens — as it has during every accelerator generation transition since 2018 — all of the accounts the router monitors lose the SKU simultaneously. The router's optimization collapses to nothing-available; the workload cannot route. The architectural problem named in MAO-P1 surfaces here in concrete form: the SKU-layer abstraction does not provide a routing escape when the upstream allocation has bound.

2.5 ACCOUNT-ONLY ROUTING INTO GOVERNANCE-ENVELOPE BREACH

An account-only router selects whichever cloud account currently has quota and price. The router does not reason about the governance envelope; it reasons about the account-level constraints exposed by each provider's API. The federation's governance envelope — sovereignty commitments, strategic-allocation positioning, regional commitments — is invisible to the account-only router; routes that would breach the envelope are not refused because the router cannot see the envelope. The Skipjack Protocol's *governance-gated execution* principle is violated structurally, not by malice but by routing-layer architectural blindness.⁷

2.6 THE UNIFIED PROBLEM STATEMENT

Each single-policy router optimizes correctly for its single dimension and fails on the other four. The unified problem is that **routing in heterogeneous compute is a five-dimensional decision** — workload class, governance envelope, substrate availability, cost regime, sovereignty constraint — and no single-policy router has the decision structure to refuse routes that exceed the envelope on any of the five dimensions. The architectural response, developed in §3, is a multi-policy router that reasons over all five dimensions, anchored by an envelope the router cannot exceed.

3. THE CAPACITY-AWARE MULTI-POLICY ROUTER

The **multi-policy router** is specified as a structured decision component composed of five evaluators — workload classifier, envelope evaluator, substrate evaluator, cost evaluator,

⁶ Yashar Ganjali et al., “GPU Scheduling for Deep Learning Workloads in Heterogeneous Environments: A Survey,” *ACM Computing Surveys* 56, no. 3 (2024): article 65, <https://doi.org/10.1145/3617995>.

⁷ Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.



sovereignty evaluator — coordinated by a routing decision engine that combines their outputs into a routing choice subject to a hard envelope constraint.

3.1 WORKLOAD CLASSIFIER

The workload classifier categorizes incoming workloads into operationally distinct classes: **steady-state inference** (sustained utilization, predictable demand), **burst inference** (elastic demand, ephemeral), **persistent memory operations** (read/write against the authoritative tier), **governance overhead** (Skipjack enforcement, comprehension functions, reconciliation), and **forward-deployed DDIL workload** (P4 cognitive nodes operating disconnected from the federation core). Each class has different routing preferences and different substrate compatibility. The classifier is the first decision point; its output is consumed by the four remaining evaluators.⁸

3.2 ENVELOPE EVALUATOR

The envelope evaluator reads the federation's governance envelope — sovereignty commitments, strategic-allocation positioning, regional commitments, persona-account assignments — and emits a hard constraint on which substrates the workload may route to. The envelope is human-defined at the governance layer (MR-P6) and updated via Skipjack sprint-planning ceremony.⁹ The evaluator does not optimize; it constrains. Routes that violate the envelope are not candidates; the evaluator's output is the set of admissible substrates, not a preference ranking.

3.3 SUBSTRATE EVALUATOR

The substrate evaluator reads runtime telemetry from each substrate — capacity availability, current accelerator allocation, persistence-tier consistency state, observability plane signals — and emits a feasibility score for each admissible substrate. The evaluator does not classify workloads (that is §3.1's job) and does not enforce the envelope (that is §3.2's job); it answers the question *given that this workload class is admissible on these substrates, which substrates can actually accept it right now?*

3.4 COST EVALUATOR

The cost evaluator reads the current cost regime for each feasible substrate and emits a per-route cost estimate. The cost estimate is workload-class-aware (steady-state on Omen amortizes capex; burst on AWS pays per-hour; sovereignty on OCI is reserved-instance; the math differs by class) and consistent with the MAO-P3 capex/opex inflection.¹⁰ The cost evaluator does not select the cheapest route as an output; it provides cost as an input to the routing decision engine, which trades cost against the other dimensions.

⁸ Christina Delimitrou and Christos Kozyrakis, "Quasar: Resource-Efficient and QoS-Aware Cluster Management," *Proceedings of ASPLOS '14*, 2014, <https://doi.org/10.1145/2541940.2541941>.

⁹ Justin H. Kuiper, CISSP, *Operating Model for Agentic Teams* (v1.0-preprint, 2026), zenodo.org/records/19869313.

¹⁰ Forthcoming MAO-P3 *The Capex/Opex Inflection* (v0.1-seed, in concurrent drafting; yks-build#94).



3.5 SOVEREIGNTY EVALUATOR

The sovereignty evaluator is, in part, redundant with the envelope evaluator — sovereignty commitments are themselves part of the envelope. It exists as a separate evaluator because sovereignty constraints can change on shorter cadences than the envelope-update cadence (a sanctions event, an export-control update, a coalition-membership change), and the routing layer must respond before the next envelope update propagates. The sovereignty evaluator is the runtime watchdog for sovereignty-related envelope violations that arrive between scheduled adjustments.

3.6 THE ROUTING DECISION ENGINE

The decision engine combines the five evaluators' outputs into a routing choice. The envelope evaluator's constraint is hard (routes that violate the envelope are excluded); the other four evaluators contribute weighted preferences over the admissible set. The weighting itself is governance-policy (different operators will weight cost vs. latency vs. capacity quality differently); the architecture admits the weighting as a policy artifact rather than hard-coding it. The engine's output is a routing decision plus an *envelope-distance* score that the comprehension instrument's envelope-distance estimation function (MR-P8) consumes.¹¹

3.7 IMPLEMENTATION: K3S + SERVICE MESH

The concrete implementation in the Robo Stack federation is built on K3s as the substrate orchestrator,¹²¹³ with a service mesh providing the substrate-spanning routing layer and a custom controller implementing the five-evaluator decision logic. Alternative implementations — Karmada for multi-cluster, Crossplane for declarative-substrate-routing, custom control planes for sovereign-jurisdiction-aware routing — are admissible by the architecture. What is invariant is the five-evaluator decision structure and the hard envelope constraint.

4. ROUTING AS A CONTINUOUS OPERATIONAL PROCESS

Routing is not a one-shot decision; it is a continuous reassessment of placement against changing substrate availability, cost regime, and envelope state. The process has three phases.

Initial routing evaluates the five dimensions at workload admission and selects the first substrate to execute on. The envelope-distance score from the initial routing is logged.

Continuous reassessment re-evaluates the five dimensions at intervals defined by the workload's characteristics. Steady-state workloads reassess on coarser cadence; burst workloads reassess every few minutes; governance-critical workloads reassess at every Skipjack reconciliation checkpoint. When the reassessment indicates that the original routing is no longer optimal — substrate availability shifted, cost regime shifted, envelope updated — the router triggers a migration decision.

¹¹ Forthcoming MR-P8 *Comprehension Instrument* (v1.0-preprint, in concurrent drafting; yks-build#98).

¹² Brendan Burns et al., "Borg, Omega, and Kubernetes," *Communications of the ACM* 59, no. 5 (2016): 50–57, <https://doi.org/10.1145/2890784>.

¹³ Klaus Rausch, Waldemar Hummer, Vinod Muthusamy, Alexander Rashed, and Shahram Dustdar, "Towards a Serverless Platform for Edge AI," *IEEE Workshop on Hot Topics in Edge Computing (HotEdge '19)*, 2019, <https://www.usenix.org/conference/hotedge19/presentation/rausch>.



Migration moves the workload to the newly preferred substrate, with explicit handling of the persistent state implications. Stateless workloads migrate freely; stateful workloads migrate with persistence-tier coordination (writes route back to the authoritative tier on Omen; the new substrate reads through the unified persistence plane).¹⁴

The three phases are operationally enforced by the orchestration plane (MR-P6's Orchestrator function) and instrumented by the observability plane (MR-P7).¹⁵¹⁶ The HGC³AE² framework's exponent — the closure — is operationally instantiated in the continuous reassessment phase: routing telemetry feeds envelope refinement which the next reassessment routes against.

5. SKIPJACK COUPLING: ROUTING UNDER ENFORCEMENT

The Skipjack Protocol's four enforcement mechanisms bind to the routing layer through the multi-policy router's five-evaluator decision logic.¹⁷

Mechanism 1 — Context integrity is enforced by the substrate evaluator's read of persistence-tier consistency state; routes that would consume stale context are surfaced (not silently routed) to the orchestrator. **Mechanism 2 — Structured task decomposition and routing** is the routing layer itself; the multi-policy router IS Mechanism 2 at federation scale. **Mechanism 3 — Human-in-the-loop control points** fire when routes approach envelope boundaries; the envelope evaluator's output is what triggers HITL when proximity to the envelope is detected. **Mechanism 4 — Iterative feedback loops** close at the Skipjack sprint-review ceremony, where routing telemetry feeds envelope refinement for the next sprint.

Each mechanism's coupling is structural: the routing layer's architecture is designed to feed the Skipjack enforcement, not to be retrofitted with enforcement after the routing is built. Routing without Skipjack coupling is what produces the failure modes of §2; routing with Skipjack coupling is what makes mission-ready operation viable across heterogeneous substrates.

6. IMPLICATIONS AND CONCLUSION

The implications of multi-policy routing extend across operating-model design, policy-as-code investment, observability-plane composition, and the broader question of how agentic systems run at scale. Three implications warrant explicit statement: routing policy is governance authority (not infrastructure team authority); the observability plane is the federation's epistemic substrate (no routing decision is better than the substrate telemetry it reasons over); and the

¹⁴ Martin Kleppmann, *Designing Data-Intensive Applications* (Sebastopol, CA: O'Reilly, 2017).

¹⁵ Justin H. Kuiper, CISSP, *Operating Model for Agentic Teams* (v1.0-preprint, 2026), zenodo.org/records/19869313.

¹⁶ Justin H. Kuiper, CISSP, *Agentic Substrate* (v1.0-preprint, 2026), zenodo.org/records/19869291.

¹⁷ Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.



routing layer is itself a security surface (a compromised router can quietly route workloads outside the envelope).¹⁸¹⁹

The core claim of this paper: **routing in heterogeneous compute is a five-dimensional, governance-anchored decision that single-policy routers cannot solve.** Operators who reason about routing as load-balancing are reasoning at the wrong layer; operators who reason about routing as multi-policy decision-making anchored by a human-defined envelope are reasoning at the layer where the actual constraint lives.

Returned to HGC³AE² letter by letter:

H — Human-governed. Routing policy is the governance layer's product. The envelope evaluator reads policy that humans authored; the router operates inside that policy and cannot exceed it.

G — Curated Context. The substrate evaluator reads curated substrate telemetry; the envelope evaluator reads curated envelope state. Curation is upstream of routing decisions.

C¹ — Cybersecurity. The router is a security surface; the routing layer's compromise compromises the federation. MAO-P9 develops the security treatment.²⁰

C² — Continuity. Workloads survive substrate failure through routing under HGC³AE²; persistence-tier coordination preserves state across migrations.

C³ — Coherence. All routing decisions evaluated against the same envelope; routing across heterogeneous substrates is coherent at the policy layer even though substrates are heterogeneous.

A — Agentically Engineered. The five-evaluator decision logic is agentic; the routing engine operates the policy that humans defined.

E — Executed. Routing runs continuously, not at admission only. Reassessment is the operational form of E for routing.

The exponent — the closure — is the routing-telemetry-to-envelope-refinement loop. The loop closes at the Skipjack sprint-review cadence.

Paper 3 (Capex/Opex Inflection) examines the cost-regime conditions under which the multi-policy router's choices favor bare-metal substrate over cloud-managed substrate, and the operational consequences of operating at and across the inflection point. The remaining MAO papers develop the operating-model implications of routing across substrates.²¹²²

18 Tim Hinrichs et al., "Open Policy Agent: A Policy Language for the Cloud-Native Era," *ACM SoCC '18 Posters*, 2018; supplementing peer-reviewed work in Andrei Sabelfeld and Andrew C. Myers, "Language-Based Information-Flow Security," *IEEE Journal on Selected Areas in Communications* 21, no. 1 (2003): 5–19, <https://doi.org/10.1109/JSAC.2002.806121>.

19 Hassan Takabi, James B. D. Joshi, and Gail-Joon Ahn, "Security and Privacy Challenges in Cloud Computing Environments," *IEEE Security & Privacy* 8, no. 6 (2010): 24–31, <https://doi.org/10.1109/MSP.2010.186>.

20 Forthcoming MAO-P9 *Security & Provenance* (v0.1-seed, in concurrent drafting; yks-build#102).

21 Forthcoming MAO-P3 *The Capex/Opex Inflection* (v0.1-seed, in concurrent drafting; yks-build#94).

22 Forthcoming MAO-P6 *Cost Guardrails* (v0.1-seed, in concurrent drafting; yks-build#100).



REFERENCES

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